

1. What is Apex?

Apex is a programming language used in Salesforce to create custom business logic and automation. It is similar to Java and runs on the Salesforce platform. Developers use Apex when Flow Builder or normal configuration is not enough for complex tasks.

2. Difference

Flow vs Apex

Flow is a no-code or low-code tool used to automate simple business processes using drag-and-drop options. Apex is coding-based and used for advanced automation and complex logic. Flow is easier for admins, while Apex is mainly used by developers.

Configuration vs Coding

Configuration means creating applications using clicks, settings, and tools without writing code. Coding means writing programs using languages like Apex to build advanced features. Configuration is faster for simple tasks, while coding gives more flexibility and control.

3. Real Examples Where Apex Is Needed

1. Automatically calculating scholarship eligibility based on multiple student conditions.
2. Sending bulk notifications to thousands of students at the same time.
3. Integrating Salesforce with external college payment systems or websites.

4. Integrated System Design

A College Management System in Salesforce can manage students, teachers, courses, attendance, and fees using CRM concepts.

Objects are used to store data like Student, Faculty, Course, and Attendance. Relationships connect these objects together. For example, one student can join many courses.

Validation rules help ensure correct data entry, such as preventing empty student IDs or invalid phone numbers.

Flows are used to automate tasks like sending admission confirmation emails or updating attendance automatically.

Apex is used for advanced functions like generating report cards, fee calculations, or connecting with external applications.

5. Pseudocode Examples

```
IF student attendance < 75%  
THEN send warning email  
IF fees are fully paid  
THEN update status as "Paid"  
WHEN new student record is created  
THEN send welcome message
```

6. Reflection

Enterprise systems eventually need programming because businesses grow and requirements become more complex. Simple tools may not handle advanced automation or integrations. Programming with Apex helps companies create customized solutions, improve performance, and manage large-scale business operations efficiently.

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Murari Naidu
Student at Vishnu institute of technology
Andhra Pradesh, India

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